

Angus Addlesee

Applied Scientist · Frontier AI Modelling Lab @
Amazon Alexa+

Cambridge, UK
✉ angus@addleseeHQ.com
🌐 www.addlesee.co.uk
in [angusaddlesee](#)
🔗 [angusaddlesee](#)

Profile

Applied Scientist in Amazon Alexa's Frontier AI Modelling Lab, working on Alexa+ across several critical workstreams: LLM routing and optimisation, reward modelling for evaluation and training, and high-throughput evaluation infrastructure. PhD in Artificial Intelligence focused on accessible, human-centred dialogue systems for people with dementia. 50+ international talks, EACL Best Demo Award, and work featured in TIME Magazine.

Experience

- 2024–Present **Applied Scientist, Amazon Alexa+ • Frontier AI Modelling Lab**, Cambridge, UK
- LLM routing and optimisation for Alexa+ (production models triggered for 100% of traffic).
 - Lead the development of reward models for model evaluation and training, moving to run on 100% of production traffic.
 - Built a high-throughput system simulator enabling offline evaluation of counterfactual tool-calling, used for router training and model checkpoint evaluation against traffic signals.
 - On-call rotas: diagnose and resolve Alexa+ production issues under pressure.
 - Active interviewer: conducted 25+ technical interviews for the team.
- 2022–2024 **Research Associate, Heriot-Watt University**, Edinburgh, UK
- EU SPRING H2020 project: conversational AI and robotics for dementia care.
 - Led the conversational AI and dialogue system development for a multi-party social robot deployed in a Parisian memory clinic.
 - EACL 2024 Best Demo Award; project featured in TIME Magazine.
- Mar–Aug 2022 **Applied Scientist Intern, Amazon Alexa**, Cambridge, UK
- Research internship during PhD on voice assistant accessibility and partial-question understanding. Submitted three papers (INTERSPEECH, CUI, AMLC).
- 2019–2022 **Machine Learning Engineer & Researcher, Wallscope**, Edinburgh, UK
- NLP and ML projects for NHS Scotland and Scottish Government (real-time discharge-letter parsing into SNOMED-grounded knowledge graphs feeding a live GP UI; child-protection analytics). Concurrent with PhD.
- 2018–2019 **Machine Learning Engineer, Wallscope**, Edinburgh, UK
- Information extraction from unstructured data into knowledge graphs for healthcare and public sector clients.
- 2017–2018 **Data Scientist, Wallscope**, Edinburgh, UK
- Data-driven NLP solutions for healthcare and government services.

Education

- 2019–2024 **PhD in Artificial Intelligence, Heriot-Watt University**, Edinburgh, UK
- Thesis: *Incremental multi-party conversational AI for people with dementia*. Funded by The Data Lab.
- 2016–2018 **MSc in Artificial Intelligence with Speech and Multimodal Interaction, Heriot-Watt University**, Edinburgh, UK, *Distinction*
- Thesis: *An Exploration into Open-domain Question Answering for Conversational Agents*.
- 2013–2016 **BSc (Hons) in Mathematics, Heriot-Watt University**, Edinburgh, UK
- Dissertation: *Markov Chains: Modelling Stochastic Processes and Exploring the Mathematics Behind these Models*.

Awards

- 2024 **EACL Best Demo Award**, European Chapter of the Association for Computational Linguistics. *Multi-party Multimodal Conversations Between Patients, Their Companions, and a Social Robot in a Hospital Memory Clinic*.
- 2022 **Societal Impact Award**, Heriot-Watt University.
- 2019 **Industrial Doctorate Funding Award**, The Data Lab Scotland.

Speaking & Leadership

- 50+ Talks International talks at venues including **Harvard, USC, KTH, BBC, NatWest**, plus conferences across Europe, North America, and the Middle East (Boulder, Boston, Dublin, Eindhoven, Glasgow, Leipzig, Los Angeles, Ljubljana, Malta, Montreal, Paris, Prague, Riga, Stockholm, Turin, Abu Dhabi, and more).
- Co-founder **European & African Conversational AI Summits** (since 2020). The first European event hosted 1,200 attendees with 46 speakers; the first African event hosted 1,000 attendees with 44 speakers, the first event of its kind in Africa.
- SIGdial Officer Special Interest Group on Discourse and Dialogue.
- Conference Organising SIGdial 2022 (Edinburgh); YRRSDS 2021, 2022; SLIDInG 7 (with the Scottish Government); ICMI 2020 (supported).
- Supervision 60+ MSc students supervised across multiple cohorts on conversational AI, accessibility, and human-computer interaction projects.

Selected Publications

- 2025 *Socially Pertinent Robots in Gerontological Healthcare*. Alameda-Pineda et al. International Journal of Social Robotics.
- 2024 *Multi-party Multimodal Conversations Between Patients, Their Companions, and a Social Robot in a Hospital Memory Clinic*. Addlesee et al. EACL (**Best Demo Award**).
- 2024 *You Have Interrupted Me Again!: Making Voice Assistants More Dementia-Friendly With Incremental Clarification*. Addlesee & Eshghi. Frontiers in Dementia.
- 2024 *Clarifying Completions: Evaluating How LLMs Respond to Incomplete Questions*. Addlesee, Lemon & Eshghi. LREC-COLING.
- 2023 *Multi-party Goal Tracking with LLMs: Comparing Pre-training, Fine-tuning, and Prompt Engineering*. Addlesee et al. SIGdial.
- 2024 **PhD Thesis: Incremental Multi-party Conversational AI for People with Dementia**. Heriot-Watt University.
Full list at Google Scholar.

Skills & Interests

- Technical LLM Routing & Optimisation, Large Language Models, Reward Modelling, Conversational AI, Real-Time Systems, Speech & Dialogue, Knowledge Graphs, Production ML.
- Research Human-AI Interaction, Accessibility, Responsible AI, Multimodal Interaction, Healthcare AI, Multi-party Dialogue.
- Languages English (native).